

Task 4 Documentation

Feature Encoding & Scaling using Adult Census Income Dataset

Introduction

Feature Encoding and Feature Scaling are crucial preprocessing techniques in Machine Learning.

Real-world datasets contain categorical and numerical variables that must be transformed before model training.

This project demonstrates Label Encoding, One-Hot Encoding, Standardization, Normalization, Data Visualization, Statistical Comparison, and Correlation Analysis.

Objective

- Understand the Adult Census Income Dataset
- Identify categorical and numerical features
- Apply Label Encoding and One-Hot Encoding
- Apply Standardization and Min-Max Normalization
- Compare distributions before and after scaling
- Prepare data for machine learning

Dataset Description

Dataset: Adult Census Income Dataset.

The dataset contains demographic and employment information used to predict whether annual income exceeds \$50,000.

Dataset Features

age, workclass, fnlwgt, education, education-num, marital-status, occupation, relationship, race, sex, capital-gain, capital-loss, hours-per-week, native-country, income.

Technologies Used

Python, Pandas, Matplotlib, NumPy, Scikit-Learn

Libraries Imported

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.preprocessing import LabelEncoder, StandardScaler, MinMaxScaler
```

Step 1: Loading the Dataset

The dataset is loaded using Pandas. Dataset structure, dimensions, and datatypes are examined using `head()`, `info()`, and `shape`.

Step 2: Identifying Categorical Features

Object datatype columns are identified as categorical variables and selected for encoding.

Step 3: Label Encoding

The target variable income is converted into numerical form:

$\leq 50K \rightarrow 0$

$> 50K \rightarrow 1$

Why?

Machine Learning models require numerical target labels.

Step 4: One-Hot Encoding

Nominal categorical variables such as workclass, education, occupation, race, and native-country are converted into binary columns.

`drop_first=True` is used to avoid the Dummy Variable Trap.

Dummy Variable Trap

When all dummy variables are retained, one variable can be predicted from the others, causing multicollinearity.

Using `drop_first=True` removes one redundant category.

Step 5: Feature Scaling

Numerical features have different ranges. Scaling ensures that features contribute equally during model training.

Step 6: Standardization

Formula:

$$z = (x - \mu) / \sigma$$

Benefits:

- Mean becomes 0
- Standard deviation becomes 1
- Useful for Logistic Regression, SVM, PCA, and KNN

Step 7: Normalization

Formula:

$$x' = (x - x_{\min}) / (x_{\max} - x_{\min})$$

Benefits:

- Values scaled to [0,1]
- Useful for Neural Networks and distance-based algorithms

Step 8: Distribution Analysis

Histograms are generated for Original, Standardized, and Normalized data to compare distributions.

Step 9: Statistical Comparison

Descriptive statistics are generated using `describe()`.

Mean, standard deviation, quartiles, minimum, and maximum values are compared.

Step 10: Correlation Analysis

A correlation matrix is generated to analyze relationships among numerical features.

Output Files Generated

age_distribution_comparison.png

correlation_matrix.png

encoded_dataset.csv

standardized_dataset.csv

normalized_dataset.csv

Learning Outcomes

- ✓ Feature Encoding
- ✓ Label Encoding
- ✓ One-Hot Encoding
- ✓ Standardization
- ✓ Normalization
- ✓ Data Visualization
- ✓ Dataset Preparation for Machine Learning

Future Improvements

- Machine Learning Model Training
- Feature Selection
- Hyperparameter Tuning
- PCA
- Model Evaluation
- Interactive Dashboard

Conclusion

Categorical features were transformed using Label Encoding and One-Hot Encoding. Numerical features were scaled using Standardization and Normalization. The resulting dataset is ready for machine learning algorithms such as Logistic Regression, KNN, SVM, Random Forest, and Neural Networks.